

1. Chunxiang is 4 years old now. His younger brother is half his age. When Chunxiang is 50 years old, how old will his younger brother be?
2. A family has two children. Given that at least one of them is a boy, what is the probability that the other child is also a boy?
3. Using exactly 6 matchsticks, what is the maximum number of equilateral triangles you can form? Remark: Maybe can prepare the matchsticks practically.
- 4.

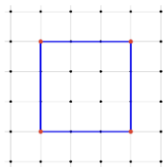


Figure 1: Square

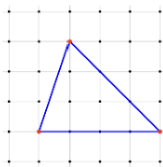


Figure 2: Triangle

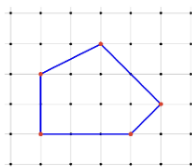


Figure 3: Pentagon

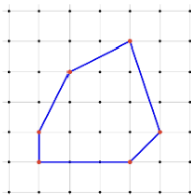


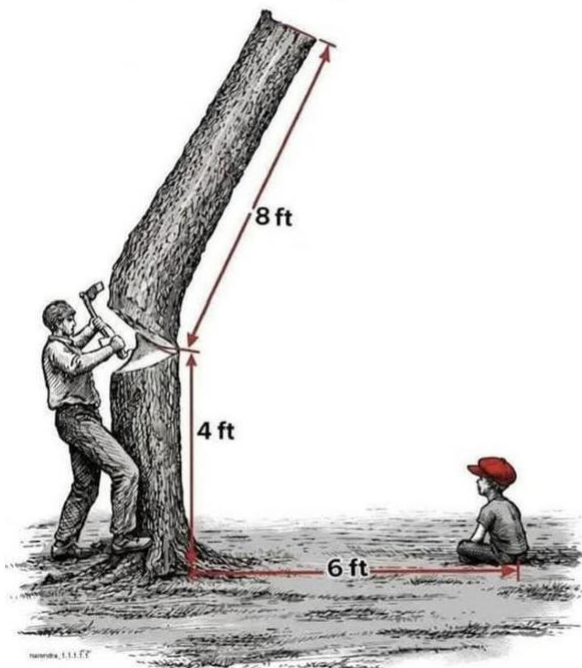
Figure 4: Irregular Hexagon

Find $Area_1 + Area_2 + Area_3 + Area_4$

Remark: Here I want to introduce the ideas of Pick's Theorem, it is quite interesting but less people know it.

- 5.

If the tree, when it falls, forms a right triangle with the ground, will the boy be safe?



Calculate that how many ft do the boy need to backward if the previous answer is no.

6.

Find $\sqrt{1.77777 \dots}$, $\sqrt{1 + \sqrt{1 + \sqrt{1 + \sqrt{1 + \dots}}}}$, $\sqrt{2 \sqrt{4 \sqrt{8 \sqrt{16 \sqrt{\dots}}}}}$

7.

EIGENSPACE

ISOMORPHISM

SURJECTION

INVOLUTION

MANIFOLD

IDEMPOTENT

STOCHASTIC

HOMOMORPHISM

DIFFEOMORPHISM

8.

$\sin^2 0^\circ + \sin^2 1^\circ + \dots + \sin^2 89^\circ + \sin^2 90^\circ = ?$
 $100 - 98 + 96 - 94 + 92 - 90 + \dots + 8 - 6 + 4 - 2 = ?$

9.

Find the Fallacy.

$$x = \frac{\pi + 3}{2} \quad (1)$$

$$2x = \pi + 3 \quad (2)$$

$$2x(\pi - 3) = (\pi + 3)(\pi - 3) \quad (3)$$

$$2x(\pi - 3) = \pi^2 - 9 \quad (4)$$

$$2\pi x - 6x = \pi^2 - 9 \quad (5)$$

$$9 - 6x = \pi^2 - 2\pi x \quad (6)$$

$$9 - 6x + x^2 = \pi^2 - 2\pi x + x^2 \quad (7)$$

$$(3 - x)^2 = (\pi - x)^2 \quad (8)$$

$$3 - x = \pi - x \quad (9)$$

$$\pi = 3 \quad (10)$$

$$a = b \quad (1)$$

$$a^2 = ab \quad (2)$$

$$a^2 + a^2 = a^2 + ab \quad (3)$$

$$2a^2 = a^2 + ab \quad (4)$$

$$2a^2 - 2ab = a^2 + ab - 2ab \quad (5)$$

$$2a^2 - 2ab = a^2 - ab \quad (6)$$

$$2(a^2 - ab) = 1(a^2 - ab) \quad (7)$$

$$2 = 1 \quad (8)$$

$$2 + 2 = 4 \quad (1)$$

$$= 4 - \frac{9}{2} + \frac{9}{2} \quad (2)$$

$$= \sqrt{\left(4 - \frac{9}{2}\right)^2} + \frac{9}{2} \quad (3)$$

$$= \sqrt{4^2 - 2(4)\left(\frac{9}{2}\right) + \left(\frac{9}{2}\right)^2} + \frac{9}{2} \quad (4)$$

$$= \sqrt{16 - 36 + \left(\frac{9}{2}\right)^2} + \frac{9}{2} \quad (5)$$

$$= \sqrt{-20 + \left(\frac{9}{2}\right)^2} + \frac{9}{2} \quad (6)$$

$$= \sqrt{25 - 45 + \left(\frac{9}{2}\right)^2} + \frac{9}{2} \quad (7)$$

$$= \sqrt{5^2 - 2(5)\left(\frac{9}{2}\right) + \left(\frac{9}{2}\right)^2} + \frac{9}{2} \quad (8)$$

$$= \sqrt{\left(5 - \frac{9}{2}\right)^2} + \frac{9}{2} \quad (9)$$

$$= 5 - \frac{9}{2} + \frac{9}{2} \quad (10)$$

$$= 5 \quad (11)$$

10. There are 9 identical-looking coins, and exactly 1 of them is fake (it has a different weight, but you do not know whether it is heavier or lighter than a real coin). Using a balance scale, what is the minimum number of weightings needed to guarantee finding the fake coin and determining whether it is heavier or lighter?

11. A palindrome poem reads the same forward and backward.

Suppose a 7-letter word follows a symmetrical pattern like **A B C D C B A** (e.g., the 1st and 7th letters are the same). If you only use the 26 standard English letters, how many different 7-letter combinations can be formed using this pattern?

12. A housing estate has 300 apartments, and each door uses a 4-digit passcode (from 0000 to 9999, so there are 10,000 possible passwords).

What is the probability that **at least two apartments share the exact same passcode**?

(Here consider the birthday problem but consider in the difference position.)

13. Nine local working-class individuals are sitting in a café in Malaysia. Each of them earns an average annual salary of **RM 60,000**. Suddenly, a top C-Suite executive earning **RM 12,000,000** per year walks in and sits down with them.

What happens to the **mean** (average) annual salary of the 10 people in the café?

Does the **median** annual salary of the 10 people change significantly?

14. In a city of 1,000 people, only **1 person** has a rare disease.

A hospital tests all 1,000 people. The test is very good, but it makes a mistake 1% of the time by telling a healthy person they are sick.

The 1 sick person gets a positive result.

Out of the 999 healthy people, 1% of them (about 10 people) also get a wrong positive result by mistake.

In total, 11 people get a positive test, but only 1 of them is actually sick!

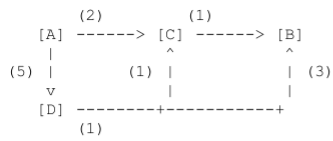
Question:

If a patient gets a positive test, what is the simplest mathematical step a doctor can take to be almost 100% sure if the patient is truly sick?

(Here I not sure about should be introduce Bayes' theorem or not)

15. A delivery robot needs to travel from **Point A (Start)** to **Point B (Destination)** through a grid of intersections.

Each road between intersections has a "traffic time" (weight) marked next to it:



Paths Available:

- Route 1: A -> C -> B
 - Time: $2 + 1 = 3$ mins
- Route 2: A -> D -> C -> B
 - Time: $5 + 1 + 1 = 7$ mins
- Route 3: A -> D -> B
 - Time: $5 + 3 = 8$ mins

Which route gives the shortest total time for the robot to reach Point B, and what is the minimum time needed?

(Here I want to introduce Dijkstra's Algorithm (or the *A Search Algorithm**))

16. Modrian Art Problem

Rules:

Complete Coverage:

You must completely fill the $N \times N$ grid with non-overlapping rectangles (squares are also allowed) without leaving any empty spaces.

Distinct Dimensions (Core Restriction):

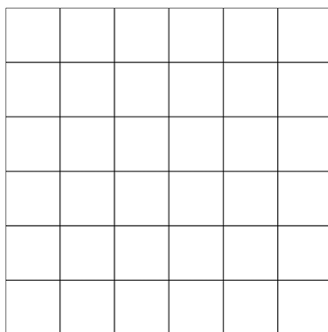
No two rectangles can have the same dimensions.

Note: A 2×3 rectangle and a 3×2 rectangle are considered to have the same dimensions (one is simply a 90° rotation of the other), so they cannot both be used in the same grid.

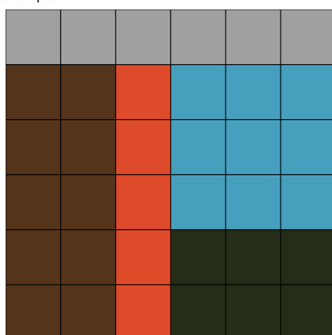
Scoring:

Calculate your total score using the formula:

Score = Area of Largest Rectangle - Area of Smallest Rectangle



example



17.

Need to fill each row with a equation such that equal to 45 (Maybe this can be a part of digital based.)

Find the equation that = 45

1 2 3 4 5 6 7 8 9 0

ENTER

+

-

×

÷

=

DELETE

?

Hint

18. Need to use the following number and operator to 24

Like $1+7=7$, then we still having 7,10,17 of number to select and $- \times /$ operation to select

TARGET **24** ?

1

10

12

6

+

-

×

÷

Reset

Undo

19. [Play Cryptograms Puzzle Online | Decode Hidden Messages](#)

(a cryptography using the probability to show each letter appear how many times.)
(Maybe this can be a part of digital based.)

20. The Monty Hall Problem

You are a contestant on a TV game show with 3 closed doors in front of you.

Behind one door is a brand-new sports car.

Behind the other two doors are goats.

Here is how the game goes:

1. You randomly pick Door 1.
2. The host, who knows what is behind all the doors, opens Door 3 to reveal a goat.
3. The host then asks you: "Do you want to stick with Door 1, or switch to Door 2?"

Question: Does switching increase your chances of winning the sports car? What is the probability of winning if you switch?

